Patient Flow Simulator — Technical Specification

EFG Apps · Whole-pathway model v7.1 · SDEC design model v3

This document sets out exactly what the two models do, so an analyst or reviewer can check the logic, reproduce results and judge whether an output is fit for a given decision.

---

1. Purpose and appropriate use

Both models are **stochastic simulations for comparing options**. You configure a baseline that matches how the hospital runs today, change one or more levers, and read the difference.

They are **not forecasts**. Each run uses a seeded random-number generator: the same inputs and seed always give identical results, and two scenarios run with the same seed face the same patients. Structural error in the assumptions therefore largely cancels when scenarios are compared, so **differences between scenarios are much more reliable than any absolute figure**.

Appropriate uses:

- ranking options and showing which constraint binds
- explaining why a downstream change moves a front-door measure
- screening options before detailed analysis

Not appropriate on their own:

- point forecasts ("ED 4-hour will be 71% next winter")
- investment-grade benefit figures for major capital decisions, which should be confirmed with a patient-level model built on local data

---

2. Architecture

Item	Detail
Stack	React 18, Recharts 2, Tailwind CSS 3, built with Vite 5

Runtime	Entirely client-side JavaScript. No server, API, database or analytics
Network	None at runtime. Fonts (Inter, JetBrains Mono, Latin subset) are bundled
Storage	Saved scenarios in the browser's `localStorage`, prefixed `pfs:`. Per user, per device, per browser
Builds	`npm run build` → `dist/` for a web host; `npm run build:single` → one self-contained HTML file
Determinism	Seeded Mulberry32 generator. Identical inputs + seed → identical output
Source	`src/FlowModel.jsx`, `src/SdecModel.jsx`: each file holds its simulation engine (pure functions, no UI dependencies) followed by its interface

---

3. Whole-pathway model

3.1 Structure

- **Level 1, front door:** ED and any other front-door units.
- **Level 2, assessment:** AMU and SDEC.
- **Level 3, inpatient wards:** grouped by division or split by specialty.
- **Exit:** discharge, with an NCTR pathway (P0–P3) for ward patients.
- **Escalation:** a capacity-capped pool of escalation beds for ward-bound patients.

3.2 Time and run length

- Hourly time steps.
- A 4-day warm-up, excluded from all results, then the chosen analysis period (7–120 days, default 28).
- Units start pre-filled so the run begins near steady state:
  - Level 1 at 40% of capacity
  - Level 3 at 90%
  - Level 2 at an estimated steady-state census (Little's law: inflow × expected stay, plus 20%, limited to 30–90%)

3.3 Order of events each hour

1. Ward discharges whose discharge time has arrived.
2. Escalation: discharges from escalation, then transfers to a home-ward bed if one is free,

otherwise to another ward if the patient has more than 12 hours' stay left.

3. Repatriation: where a home ward has a free bed, one of its outlying patients moves back, longest remaining stay first, provided they have more than 12 hours left. At most one per ward per hour.
4. Level 2, then Level 1, patients who are ready attempt their move (rules in 3.6).
5. New arrivals join the queue at their entry unit.
6. Queues admit into free capacity, deepest level first.
7. Statistics are recorded.

The order is a priority rule: beds freed in step 1 go first to escalation boarders, then to internal transfers, then to new arrivals.

3.4 Arrivals

- Each source has a daily rate and a pattern through the day: a typical ED attendance curve, or flat.
- Arrivals per hour are Poisson with mean `rate / 24 × hourly factor`. The hourly factors average 1.0.

3.5 Routing and acuity

- When a patient enters a Level 1 or 2 unit, their onward destination is drawn once from the unit's routing percentages.
- Each source's acuity multiplier scales those percentages, capped at 95% in total. Whatever is left is discharge home.
  - Example: a 25% admitted share × ambulance acuity 2.0 gives about 50% ambulance conversion.
- Time in a Level 1 or 2 unit is drawn from an exponential distribution whose mean depends on the destination (LOS if discharged, or LOS per onward route).
- Enter **clinical** time only. Waits for downstream beds are produced by the model.

3.6 Moves, blocking and pressure valves

- A ready patient whose destination is full stays in their current bed. This is blocking, and it is how pressure travels upstream (ward → AMU → ED → ambulance queue).
- A patient whose **destination is a ward**, held past the outlier trigger (default 8 hours):
  1. outlies to the ward with most free beds, keeping their home-specialty LOS and NCTR profile, plus the displacement LOS penalty (default 1.0 day); or
  2. if no ward bed exists, boards in escalation (same penalty) while escalation capacity remains; or

3. otherwise stays blocked.

- A patient bound for **AMU** waits in ED. Someone needing assessment cannot be outlied to a ward, so this wait lands on ED performance.
- All statistics for outliers and boarders are attributed to their **home** specialty.

3.7 Ward stays and discharge

Ward stay = core LOS + displacement penalty (if any) + NCTR pathway delay + wait for a discharge slot.

- **Core LOS:** exponential, around each ward's mean.
- **NCTR pathway:** drawn at admission. P1–P3 shares are the site-level percentages × the ward's complexity multiplier (capped at 95% in total); P0 takes the rest. Each pathway delay is exponential around its configured mean.
- **Discharge-time curve:** two inputs (% discharged by midday and by 5pm) set the share of discharges in the three parts of the day (00–12, 12–17, 17–24), each with a fixed shape inside it. A ready patient's discharge hour is drawn from this curve.
- **Weekly pattern:** each day has a discharge propensity.

Day	Classic	Flat	Friday peak
Mon	1.0	1.0	0.80
Tue	1.0	1.0	0.90
Wed	1.0	1.0	1.00
Thu	1.0	1.0	1.10
Fri	1.0	1.0	1.20
Sat, Sun	weekend rate (default 0.60)	1.0	weekend rate

A discharge landing on a day with propensity below 1 is moved to 09:00–15:00 the next day with probability `1 - propensity`, repeated for at most 4 days.

3.8 Measures

Measure	Definition
ED 4-hour	% of patients leaving any Level 1 unit whose time from joining its queue to leaving (home or transfer) is ≤ 4h

ED 12-hour	% of the same patients with that time > 12h
Ambulance handover	% of ambulance arrivals waiting ≥ 1h in the queue before entering ED; mean wait also shown
Escalation	Mean and peak escalation beds in use; % of hours with any in use
NCTR census	Mean number of ward and escalation patients past clinical readiness
Ready → discharge	Mean time from clinical readiness to discharge
Admitted LOS	Mean ward stay, ward admission to discharge, home specialty
Ward occupancy	Bed-weighted mean occupancy of Level 3 (escalation excluded)
Bottleneck score	Bed-days per day a unit blocks or displaces: patients held upstream waiting for it, its outliers, and its boarders
OPEL (indicative)	Heuristic band from ward occupancy, handover delays, outliers and escalation use. Not an official OPEL calculation

3.9 Presets

- Small, medium and large sizes, each with divisional or specialty ward layouts.
- Preset routing is derived so each preset starts balanced at about 95% ward utilisation: `beds × 0.95 ÷ (core LOS + multiplier × 2.56 + 0.45)`.
- **Presets are illustrative and not calibrated to any real organisation.** Replace them with local figures before use.

---

4. SDEC design model

4.1 Time and run length

- 5-minute time steps.
- A 1-day warm-up, then the chosen period (default 28 days).

4.2 Patients

- Each stream generates Poisson arrivals from three sources:

    - **ED-streamed:** ED curve; weekends at 90% of weekdays
    - **GP / direct:** concentrated 09:00–18:00; weekends at 35%
    - **Ambulance direct:** ED curve; weekends at 95%

- **Same patients in every scenario.** Each stream has its own random-number stream, and every patient makes exactly six draws when they arrive: streamed or not, trolley or chair, review length, test length, outcome, held or not. Their whole journey is fixed then. A scenario can only change *where and when* a patient is seen, never *who* arrives or *what care they need*. A surge in one stream cannot change another stream's patients.

4.3 Rules every patient follows

1. **Streaming.** ED patients are sent to SDEC at the location's streaming rate and arrive after the transfer time. GP and ambulance-direct patients are not affected by streaming.
2. **Out of hours.** If SDEC is closed or past last acceptance, the configured share are held (kept in ED, or asked to return) until the next acceptance time, as long as that is within the hold limit. The rest go to ED/AMU. Held patients all arrive at opening.
3. **Spaces.** Patients queue first come, first served. They use their own stream's space first, then the shared pool. Trolley patients need a trolley; chair patients take a chair, or a trolley if no chair is free.
4. **Turned away.** A patient with no space within the turn-away time (default 2 hours) goes to ED/AMU.
5. **Clinician.** In a space, patients wait for a clinician from their stream (or any clinician, if the rota is pooled), longest wait first. **This wait is an output.**
6. **Review and tests.** The review lasts its configured time (Erlang-2). The patient then keeps the space for tests and results (Erlang-3), with separate means for patients who will be discharged and those who will be admitted.
7. **Closing.** Patients still queuing count as turned away. Patients on the unit transfer to AMU, or the unit stays open past closing if that option is chosen.

Erlang distributions are used because they are less long-tailed than an exponential and fit assessment and turnaround times better.

4.4 Layouts

- **Separate:** each stream has only its own spaces.
- **Co-located:** each stream protects a set percentage of its own spaces ("Protected when co-located"); everything else goes into one shared pool.
    - 0% protected means fully pooled.
    - 100% protected reproduces separate units exactly, and this is checked.

4.5 What each run produces automatically

Analysis	What it does
Layout comparison	Separate, fully pooled and ring-fenced on identical patients. If no ring-fence is set, a trial setting protects 50% of every stream except the surging one
Bleed check	Per stream: patients seen, turned away for lack of space, % of open time with no trolley free, wait for space, wait to be seen, most spaces in use at once
Surge test	The largest stream (or a chosen one) at +25% and +50% demand, with all other streams' patients unchanged
Size sweep	50–150% of current spaces, separate and pooled
Hours	Closing times 16:00–24:00, plus 24 hours
Robustness	Pooled minus separate across five random seeds
Holding	Compared with a run in which no out-of-hours patients are held

4.6 Value

- Admissions avoided per day = same-day discharges × the share assumed to be avoided admissions (default 60%).
- Bed-days saved = admissions avoided × avoided LOS (default 1.5 days).
- Beds equivalent = bed-days ÷ 0.92.

This value depends heavily on the avoided-admission assumption, which should be evidenced locally.

4.7 Sizing rule of thumb (shown for comparison only)

- Average spaces in use = patients per open hour × hours on the unit.
- Planning need = average + 1.5 × $\sqrt{}$average.
- Separate units each need their own allowance for peaks; one pooled unit needs one. This is why pooling needs fewer spaces.

---

5. Verification

5.1 Checks shown in the app on every SDEC run

Check	Expectation
Eligible = seen + turned away (closed) + turned away (no space) + not streamed + still waiting	Exact
Seen = discharged + admitted + transferred at close + still on unit	Exact
Space-hours counted from spaces = space-hours counted from patients	Exact
Simulation vs hand calculation (eligible, closed, held, not streamed, reached SDEC)	Within random variation
Review and test times realised vs entered	Within random variation

The hand calculation applies the same opening, streaming and holding rules arithmetically over an average week in 5-minute slots, with no random numbers.

5.2 Tests run before release

SDEC model

- Patient accounting balanced exactly in 60 configuration and seed combinations.
- 18 edge cases ran without errors or invalid numbers: blanks, zeros, no clinicians, no streams, impossible hours, extreme demand, out-of-range percentages.
- Same seed gives identical output. Eligible patients are identical across layouts, sizes, hours and locations, and a surge in one stream leaves the others' patients unchanged.
- 100% ring-fence reproduces separate units exactly.
- Hand calculation vs mean of 20 seeds × 56 days agreed within 1.5% on eligible and reached patients across six configurations. Example (default inputs): held 9.6 by hand vs 9.5 simulated per day.
- Patients seen rise with size, hours and streaming rate.

Whole-pathway model

- Behavioural tests: halving NCTR delays lowered occupancy and escalation; more ward beds lowered occupancy; +10% demand worsened ED and handover; same seed gives identical output.

Both

- Browser tests at desktop and phone width: no console errors, no external network requests, fonts loaded from the bundle.

5.3 Calibration and reporting before use

1. **Back-test.** Set the model to how you run today and compare it with your actual figures. In SDEC, enter actual seen per day on each stream card; the model should be within 10%.
2. **Calibrate inputs, not outputs.** Tune demand, capacity, LOS and NCTR. ED waits, occupancy and escalation are results.
3. **Replicate.** Run at least 10 seeds per scenario and report the median and range.
4. **Test sensitivity.** Vary the least-certain inputs: displacement penalty (0.5–1.5 days), outlier trigger (4–12 hours), demand (±10%), NCTR delays (±20%), SDEC test times and the avoided-admission share. If the ranking of options changes, say so.
5. **Keep an audit trail.** Store the exported configuration with every reported result. It contains every input including the seed.

---

6. Known limitations

- Lengths of stay are statistical distributions, not empirical patient-level data. The exponential ward LOS understates the very-long-stay tail.
- Whole-pathway time steps are hourly, so the 4-hour measure has ±30-minute resolution.
- No elective stream, seasonal variation or readmissions.
- Conversion rates stay fixed when the hospital is crowded.
- Outlying goes to the emptiest ward regardless of clinical suitability.
- Escalation and staffing costs are not modelled; escalation is priced only through its LOS penalty.
- SDEC diagnostic turnaround is an input and does not lengthen when the unit is busy, which may flatter pooling slightly.
- SDEC patients are seen first come, first served, with no clinical priority.
- Time spent in ED by patients held for SDEC is reported but not passed automatically into the whole-pathway model.
- The OPEL band is indicative only.

---

7. Data and privacy

- The application holds no patient-identifiable data and collects none.
- All inputs are aggregate operational assumptions entered by the user, and processing

happens on the user's device.

- Saved scenarios stay in that user's browser.
- When hosted publicly, the source code and the default preset values are visible to anyone. Check that defaults contain nothing your organisation would not publish.